

# TOMÁS ELÍAS VALDERRAMA PLATZ

Project Engineer | Data Analyst | M.Sc. in Geophysics

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## PROFESSIONAL PROFILE

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Data analyst and project engineer with a background in geophysics and experience in processing environmental, oceanographic, and satellite data. Proficient in Python, statistical analysis, and data visualization, with a proven ability to automate workflows and build data pipelines. Focused on turning complex data into actionable insights for technical, academic, and regulatory organizations.

## TECHNICAL SKILLS

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**Advanced:** Python (pandas, numpy, matplotlib, cartopy, geopandas, shapely, xarray, netCDF4), MATLAB, Bash, Git/GitHub, Excel/Google Sheets, ArcGIS, QGIS

**Intermediate:** Google Earth Engine (JavaScript), React/JavaScript (HTML/CSS), R

**Numerical Models:** CROCO-ROMS, OpenDrift, ILLUMINA, WRF, SWAN (learning)

## PROFESSIONAL EXPERIENCE

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### Project Engineer - Pelícanos SpA

*Dec 2025 – Present*

*Environmental engineering and applied oceanography.*

- Processing and analysis of wind, marine current, dissolved oxygen, and salinity data; preparation of technical reports in compliance with NCh 1821-2020.
- Development of Python code to automate analysis workflows and generate maps with ESRI satellite imagery, replacing a manual process in Global Mapper and PowerPoint.
- Preparation of wind reports (1-month and 1-year), current reports (1-month and 6-month), and oceanographic equipment reports for dissolved oxygen and salinity.
- Participation in field campaigns to retrieve data from wind and water quality instruments.

### Data Science and Remote Sensing Consultant

*Jul 2025 – Present*

*Project-based work under direct contracts and via LATLON SpA.*

- *Satellite pipeline (Jul–Aug 2025; client: FONDART researchers, U. de Chile):* Design and implementation of a bulk Sentinel-2 and MODIS image download workflow via Google Earth Engine (JavaScript/GEE); 3-year coverage for the Valparaíso, Metropolitan, and O'Higgins regions; delivery of documented code for project continuity.
- *Light pollution model (Oct–Dec 2025; clients: AURA and OPCC):* Installation and configuration of the ILLUMINA model; preparation of an installation manual; design and execution of *benchmark* experiments using Hawaii data; definition of input/output structure and reference figure for a night-sky monitoring web platform in northern Chile.
- *Scientific Advisor — sup3rwind / Exploradores de Energía Project, Ministry of Energy (contract with U. de Los Lagos, Jan–Dec 2026):* Scientific validation of sup3rwind (NREL), a GAN-based model downscaling ERA5 to 2 km / 5 min for South America (2005–2024), to inform its adoption as a reference wind dataset in the Exploradores de Energía platform. Three-level comparison framework (sup3r vs. in-situ, WRF vs. in-situ, sup3r vs. WRF) with bias, RMSE, Weibull ( $k$ ,  $c$ ) and P90/P99 percentile metrics; programmatic data access via NREL HSDS API (h5pyd). Advisory on infrastructure and protocols for hosting the dataset in Chile.
- *Meteorological study — Altos del Sol - Monte Mina transmission line (client: Agrosonda, Apr – May 2026):* Extreme wind and temperature analysis (Gumbel, 50-year return period, IEC 60826) for structural tower design; ice sleeve thickness estimation with the Goodwin v4 model. Station-level quality control: detection of instrumental cap and removal of unrepresentative series overlooked

in the reference study. Data sources: API Exploradores de Energía and ERA5-Land; reproducible code for similar studies.

### Graduate Research Assistant

2021 – 2025

*Universidad de Concepción — FONDECYT Regular N°1211230: “Modelling the role of biogeochemical river input into fjord systems in Chilean Patagonia”*

- Analysis of the semi-permanent eddy dynamics in the Gulf of Ancud using CROCO-ROMS ocean model outputs: vertical structure, temporal variability, and tidal role.
- Lagrangian particle transport simulations with OpenDrift at multiple depths (0–250 m), with and without tidal forcing.
- Degree awarded (2025). Thesis-derived manuscript in preparation for submission to an indexed international journal.

### Internship - IFOP (Instituto de Fomento Pesquero)

2020

*Castro, Chiloé, Chile*

- Comparative analysis of streamflow from the VIC hydrological model against 33 freshwater sources in the MOSA-ROMS oceanographic model, covering fjords and channels in southern Chile (Los Lagos and Aysén regions).
- Identification of rivers to update and new sources to incorporate into MOSA-ROMS, resulting in a net increase of  $\sim 2,800 \text{ m}^3/\text{s}$  of freshwater input to the system; work integrated into the project's final report (IFOP, 2020).
- Processing and quality control of oceanographic and environmental data in MATLAB.

## EDUCATION

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**M.Sc. in Geophysics** (Modeling and Data Analysis)

2021 – 2025

Universidad de Concepción, Chile

**B.Sc. in Geophysics**

2015 – 2021

Universidad de Concepción, Chile

## SELECTED PROJECTS

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- **Monitoring Map Generator** — Python tool ( $\sim 1,500$  lines) that integrates KMZ, shapefiles, and JSON to generate multi-page cartographic maps (JPG/PDF) with equipment auto-discovery and ERA5 overlay. Replaced a manual workflow using three tools (Global Mapper + PowerPoint).
- **AFP Fund Evaluator** (live demo) — React/Recharts web app comparing historical performance of pension funds A–E with savings projections, Monte Carlo simulation, and Sharpe ratio. Data from 2002–2024 validated against Superintendencia de Pensiones records.
- **GEE Satellite Pipeline** — Reproducible bulk download workflow for Sentinel-2 and MODIS images via Google Earth Engine; multi-year spatial coverage analysis for three Chilean regions with MGRS tiling in Python.
- **ILLUMINA** — Installation, configuration, and benchmarking of the radiative transfer model for light pollution monitoring; northern Chile.
- **Freshwater inputs to MOSA-ROMS** — Comparative analysis of VIC streamflow vs. oceanographic model sources for fjords in southern Chile; results integrated into IFOP technical report P-656118 (2020).

## LANGUAGES

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- English: Advanced — technical reading, writing, and oral comprehension
- Spanish: Native